

Vidush Singhal

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📄 [vidush-singhal](#) | 🌐 [vidsinghal](#) | 🎓 [Google Scholar](#) |

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RESEARCH INTERESTS

I am a Ph.D. student in ECE at Purdue working on compiler techniques for performing program transformations using static analysis and whole program analysis. My research interests include compilers, data layout optimization, SIMD vectorization, memory correctness/type safety, automatic parallelization, compilers for GPUs, compilers for AI/ML, compilers for QC, PL and HPC. I have experience building compiler infrastructure in LLVM, Clang, MLIR and Gibbon.

EDUCATION

- **Purdue University**, West Lafayette, IN, USA
Direct Ph.D. in Electrical and Computer Engineering May 2021 – May 2027
Advisor: Prof. Milind Kulkarni
M.S. in Electrical and Computer Engineering (en route to Ph.D.)
Non-thesis, course-only
B.S. in Computer Engineering August 2017 – May 2021

EXPERIENCE

- **Purdue University** [[📄](#)] May 2021 – Present
Graduate Research Assistant; Advisor: Prof. Milind Kulkarni West Lafayette, IN, USA
 - **Gibbon Compiler**
 - * Designed and implemented compiler passes for optimizing the layout of recursive datatypes.
 - * Enhanced runtime performance of programs by improving access patterns and spatial locality.
 - * Measured speedups of **1.14x to 54x** over the best prior work.
 - * Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.
 - * Structure-of-arrays transformation enables selective traversal, our benchmarks saw speedups of **1.1x to 12x**.
 - * Designed optimizations such as, recursion to loopification, sharing and vectorization (speedups of **1.1x to 21.4x**).
 - * Developing a **formal language** to write well-typed SoA programs and prove **memory correctness** (type safety).
 - * **Researched** on Gibbon's potential to accelerate ML inference and its execution on ML hardware such as GPUs.
 - * Researching GPU execution of **map** traversals over packed algebraic datatypes.
 - **Orchard Compiler**
 - * Developed a compiler framework to automatically fuse and parallelize tree traversals.
 - * Measured speedups ranging from **1x to 5x** over baseline tree traversals.
 - **Cornucopia**
 - * Developed a fuzzing-based compiler framework to generate a corpus of diverse binaries from source programs.
 - * Used compiler optimization flags (11c) optimization flags to drive differences in binaries.
 - * Discovered bugs in LLVM compiler optimization passes.
 - * Exposed decompilation failures in angr, Ghidra, and related binary analysis tools on generated binaries.
 - * Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools (BATs).
 - **Copse Compiler**
 - * Developed a compiler for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.
- **Microsoft Research** [[🌐](#)] May 2025 – August 2025
Compilers Research Intern; Mentor: Dr. Angélica Moreira Redmond, WA, USA
 - Developed **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
 - Implemented a **Clang-to-Clang** source transformer (~7K LOC of C++) that emits Pulse code for F* verification.
 - Built a C benchmark suite and validated generated verification conditions on representative annotated C programs.
 - Implemented and verified components of DPE (Dice Protection Environment) written in C through C2Pulse.
- **Lawrence Livermore National Laboratory** [[🌐](#)] May 2024 – August 2024
Computing Scholar Intern; Mentor: Dr. Johannes Doerfert Livermore, CA, USA
 - Engineered an LLVM Attributor pass to **optimize byte layout within stack allocations** for improved spatial locality.
 - Implemented **LLVM IR** analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
 - Optimized the **LLVM GPU sanitizer** by hoisting redundant checks out of loops, reducing XSBench runtime by **6%**.
 - Built an LLVM IR basic-block permutation pass to evaluate performance sensitivity to code layout.

• Nod.ai Labs [🌐]

Machine Learning Compiler Engineering Intern; Supervisor: Anush Elangovan

May 2022 – August 2022

Santa Clara, CA, USA

- Contributed to **Torch-MLIR**, a compiler stack for lowering programs from the **PyTorch ecosystem to MLIR**.
- Implemented kernels and compiler support for multiple **PyTorch operators** in the Torch-MLIR pipeline.
- Added support for the **Facebook DLRM** recommendation model in the PyTorch-to-MLIR compilation pipeline.
- Investigated improvements to the **ILP solver** used by the Alpa scheduling framework.

• Senior Design Project, SoCET Team [🏢]

B.S. Final Project; advised by Dr. Mark Johnson and Prof. Samuel Midkiff

July 2020 – May 2021

West Lafayette, IN, USA

- Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
- Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
- Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

• Astrome Technologies Private Limited [🌐]

Engineering Intern; Supervisor: Dr. Prasad Bhat

May 2019 – August 2019

Bengaluru, Karnataka, India

- Integrated an eMMC IP block with an AXI-Lite-to-Wishbone bridge using a SystemVerilog wrapper.
- Verified the eMMC IP and bridge integration through Vivado behavioral simulations.
- Built XSLT and Python tooling to render XML data as internal HTML reports.
- Implemented convolutional encoding and Viterbi decoding algorithms in C for error correction and detection.
- Achieved an **order-of-magnitude speedup** over the original Python implementation.

SELECTED PROJECTS

• Global Tensor Layout Optimization for IREE

Researched tensor layout optimization in the `iree-global-optimization-opt-level` compiler pipeline.

• LLM-Assisted Specification Inference for Verus

[Explored](#) LLM-assisted inference of Verus specifications using symbolic reasoning tools like Seahorn and Crucible.

• Explored OpenMP GPU Offloading for Parallel STL-style Algorithms

[Tested](#) a proof-of-concept evaluation of OpenMP GPU offloading for STL-style handwritten algorithms to validate speedups.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, T=THESIS

- [P.1] **Vidush Singhal**, Mikah Kainen, Artem Pelenitsyn, Michael H. Borkowski, Mike Vollmer, Milind Kulkarni "SoCal: A Language for Memory-Layout Factorization of Recursive Datatypes" preprint under submission, 2026 [\[ArXiv\]](#)
- [C.1] Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni "Vectorized Secure Evaluation of Decision Forests" in PLDI 2021 [\[ACM DL\]](#)
- [C.2] **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry "Cornucopia: A Framework for Feedback Guided Generation of Binaries" in ASE 2022 [\[ACM DL\]](#)
- [C.3] **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni "Optimizing Layout of Recursive Datatypes with Marmoset" in ECOOP 2024 [\[LIPICs\]](#)
- [J.1] **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni "Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals" in TACO 2024 [\[ACM DL\]](#)
- [C.4] Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton "Garbage Collection for Mostly Serialized Heaps" in ISMM 2024 [\[ACM DL\]](#)
- [C.5] Michael Rainey, Michael H. Borkowski, Michael Vollmer, Chaitanya S. Koparkar, Mikah Kainen, **Vidush Singhal** "LoCalMem: Type-Directed Adaptive Serialization for Location- and Content-Addressable Memory" in ICFP 2026 [\[ACM DL\]](#)

SELECTED COURSEWORK

Compilers	Compiler Code Generation and Optimization · Compilers for GPUs · Introduction to Compilers and Translation Engineering
Programming Languages	Reasoning About Programs · Programming Languages · Advanced C Programming · Object-Oriented Programming in C++
Algorithms & Theory	Computation Models and Methods · Applied Algorithms · Data Structures in C · Formal Languages, Computability, and Parallelization
Systems	Computer Architecture · Programming Parallel Machines · Microprocessors and Microcontrollers · Digital Design
Mathematics	Discrete Mathematics · Graph Theory · Category Theory · Random Variables
AI & Data Science	Introduction to Artificial Intelligence · Python for Data Science
Security & Quantum	Holistic Software Security · Introduction to Quantum Computing

SKILLS

Programming Languages

Experienced: C, C++, Haskell, Python

Familiar: Java, Bash, CUDA, LLVM-IR, MLIR, MATLAB, Assembly (ARM, x86), SystemVerilog

Compiler & Verification	LLVM, GCC, Coq, Dafny, Pulse, GDB
Parallel & HPC	OpenMP, MPI, Pthreads, OpenCilk, Intel TBB, Intel Cilk Tools
Data & Machine Learning	PyTorch, TensorFlow, NumPy
Systems & Databases	Unix/Linux, Windows, PostgreSQL, AFL++, gem5
Web & Development Tools	HTML, CSS, Git, Visual Studio Code, Vim, CLion, Notepad++, Kate
Professional Skills	Communication, Teamwork, Inclusive collaboration, Motivating peers

TEACHING EXPERIENCE

• Teaching Assistant, Prof. Milind Kulkarni, Intro to Data Science, Purdue University	Fall 2025
• Teaching Assistant, Prof. Milind Kulkarni, Intro to Data Science, Purdue University	Spring 2023
• Undergraduate Teaching Assistant, Prof. Cheng Kok Koh, ECE 368, Purdue University	Spring 2020
• Student Grader for Signals and Systems, Prof. Chih-Chun Wang, ECE 301, Purdue University	Spring 2020

TALKS GIVEN

• MWPLS 2025 , AoS To SoA Transformation of packed ADTs	Fall 2025
• MWPLS 2024 , Optimizing Layout of Recursive Datatypes with Marmoset	Fall 2024
• ECOOP 2024 , Optimizing Layout of Recursive Datatypes with Marmoset	Fall 2024
• PurPL Seminar , Optimizing Layout of Recursive Datatypes with marmoset	Spring 2024
• ASE 2022 , Cornucopia: A Framework for Feedback Guided Generation of Binaries	Fall 2022

HONORS AND AWARDS

• Received a travel grant, LLVM developers meeting	October 2024
• Semester Honors, Purdue University	Fall 2019 - Spring 2020
• Deans's List and Semester Honors, Purdue University	Fall 2017 - Spring 2018

LEADERSHIP EXPERIENCE

• Seminar Co-Coordinator, Purdue Programming Languages and Systems Research Group (PurPL)	Fall 2025 - present
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REVIEW EXPERIENCE

• Member of the Artifact Evaluation Committee (AEC), CGO 2027	Fall 2026
• Member of the Artifact Evaluation Committee (AEC), ASPLOS 2027	Fall 2026
• Member of the Artifact Evaluation Committee (AEC), PACT 2026	Fall 2026
• Member of the Artifact Evaluation Committee (AEC), ICFP 2026	Summer 2026
• Member of the Program Committee (PC Member), C3PO 2026	Spring 2026
• Member of the Artifact Evaluation Committee (AEC), PLDI 2026	Spring 2026
• Member of the Artifact Evaluation Committee (AEC), SPLASH 2026	Spring 2026
• Additional Reviewer, PPoPP 2026	Fall 2025
• Member of the Reviews for the Journal of Open Source Software, JOSS	Fall 2025
• Member of the Artifact Evaluation Committee (AEC), POPL 2026	Fall 2025
• Member of the Artifact Evaluation Committee (AEC), CGO 2026	Fall 2025
• Member of the Artifact Evaluation Committee (AEC), ICFP 2025	Summer 2025
• Member of the Artifact Evaluation Committee (AEC), PLDI 2025	Spring 2025
• Collaborative Reviewer with Advisor (Milind Kulkarni), CGO 2025	Spring 2025
• Member of the Artifact Evaluation Committee (AEC), CGO 2025	Fall 2024
• Member of the Artifact Evaluation Committee (AEC), PPoPP 2024	Spring 2024
• Member of the Artifact Evaluation Committee (AEC), PPoPP 2023	Spring 2023

VOLUNTEER EXPERIENCE

• Talk moderator, LLVM developers meeting 2024	Fall 2024
• Student Volunteer, ECOOP/ISSTA 2024	Spring 2024
• Student Volunteer, SPLASH 2021	Fall 2021

PROFESSIONAL MEMBERSHIPS

• Eta Kappa Nu, Purdue Electrical and Computer Engineering Honors Society	Fall 2018 - Present
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MENTORING

• Logan Anderson, Phd @ Purdue	Fall 2025
• Aniruddh Srivastava, Undergrad @ Purdue	Fall 2025
• Peter A Kaya Gretchikha, Undergrad @ Purdue	Fall 2024 - Fall 2025
• Mikah Kainen, Undergrad @ Purdue	Fall 2023
• Qingyuan (Peter) Li, Undergrad @ Purdue, SURF summer research mentor	Spring 2023 - Fall 2023